Kristin "Kay" Rubio

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Full name: Kristin Sweeney Rubio
Full name before 2020: Kristin Kelly Sweeney

Digital Portfolio
LinkedIn
GitHub
(full links below for printed CV)

SUMMARY

I am a lifelong learner with an interdisciplinary background in computer science and psychology who is passionate about language, communication, and social connection in humans and other species. For over 4 years, I have worked as a software engineer at Curriculum Associates, an educational technology company where I develop reading lessons and speech recognition technology. Prior to that, I have data science experience in medical and psychology research. In my free time, I record wildlife, analyze sound, and learn about the emerging ways that machine learning and bioacoustics can support education, animal welfare, and environmental conservation.

DEGREES

Bachelor of Science in Computer Science September 2019 – August 2022 (Completed)
Framingham State University, Framingham, MA, United States
GPA: 3.95/4.00 | Outstanding Computer Science Senior Award

Master of Arts in Psychology September 2014 – May 2016 (Completed)
Clark University, Worcester, MA, United States
(program does not issue letter grades) | Research and teaching assistantship
Thesis: Sweeney, K. K., Goldberg, A. E., Garcia, R. L. (2017). Not a "mom thing": Predictors of gatekeeping in same-sex and heterosexual parent families. *Journal of Family Psychology, 31*(5), 521-531. https://doi.org/10.1037/fam0000261
Role: I gathered and analyzed data on communication between parents in diverse families. I explored predictors of unhealthy communication patterns using regression and multilevel modeling.
Supervisor: Abbie Goldberg

Master of Science in Mental Health Counseling September 2012 – May 2014 (Completed)
University of Massachusetts Boston, Boston, MA, United States
GPA: 4.00/4.00 | Research assistantship

Bachelor of Arts in Social Studies September 2006 – December 2009 (Completed)
Ithaca College, Ithaca, NY, United States
GPA: 3.74/4.00 | Presidential Scholarship

Portfolio: https://kaysrubio.github.io/digital-portfolio/
LinkedIn: https://www.linkedin.com/in/kay-rubio-731abb114/
GitHub: https://github.com/KaySRubio

EMPLOYMENT HISTORY

Software Engineer II January 2022 – present (full-time)
Curriculum Associates, Lesson Development and AI Innovation teams
North Billerica, MA, United States

- Develop React-based web applications that host lessons for students in grades K-8
- Ensure web technologies are accessible to users who are blind, deaf, or have other disabilities
- Explore, develop and implement Python-based technologies for speech recognition pipelines
- Fine-tune neural networks for audio classification tasks

Senior Research Data Manager October 2018 – December 2021 (full-time)
HealthCore & Anthem Inc., Department of Clinical Research
Watertown, MA, United States

- Built one of the largest databases to study the impact of COVID-19 on children in North America
- Coded programs in SAS/SQL to identify data issues and improve data cleaning pipelines
- Led meetings to present progress with scientists, statisticians, and other stakeholders

Note: During my time here, I also started a second bachelor's in computer science to enhance my programming skills, which led a career transition into software engineering

Clinical Research Coordinator II May 2016 – October 2018 (full-time)
Massachusetts General Hospital, Department of Neurology
Boston, MA, United States

- Coordinated research on adults with neurological conditions affecting speech and language
- Designed research database systems, cleaned and analyzed data
- Studied the ways that changes in communication affect family relationships and wellbeing

Research Coordinator & Teaching Assistant September 2014 – December 2015 (part-time)
Clark University, Department of Psychology
Worcester, MA, United States

- Managed longitudinal studies and coordinated data collection in research on social relationships
- Screened and analyzed data and co-authored publications
- Taught lectures, led discussions and graded homework as a teaching assistant for 3 semesters

Research Assistant & Lab Manager September 2012 – May 2014 (part-time)
University of Massachusetts Boston, Department of Counseling and School Psychology
Boston, MA, United States

- Conducted research on social relationships and wellbeing among diverse groups
- Cleaned and analyzed data using t-tests and multiple regression
- Managed the research team website, social media sites, and email listserv

Research Assistant November 2011 – July 2012 (full-time)
Brigham and Women's Hospital, Department of Cardiovascular Medicine
Quincy, MA, United States

- Oversaw data collection for multi-site clinical drug trials
- Assisted research site coordinators with using the data collection software

RECENT PROGRAMMING PROJECTS

Chicken Vocalizations August 2026

Personal project

I have a small flock of chickens and I'm curious about how machine learning can reveal meaning and structure behind animal vocalizations. Inspired by Dr. Suresh Neethirajan's research, I recorded my hens doing various activities and coded python programs to explore the data. I used voice activity detection to separate recordings into individual calls, then extracted audio features. I used principal component analysis to reduce dimensionality in the data and K-means clustering to fit the calls into 8 acoustic clusters. I would be very interested in continuing this work to learn more about animal communication and support animal welfare. For more information, see my Chicken Vocalization project page.

Age Classifier March 2026

Curriculum Associates

I recently completed a work project fine-tuning a neural network. I wrote a data preparation program to assemble 20 hours of speech data. Then I used HuggingFace methods to add a classification head to a pretrained automatic speech recognition model. Using Amazon Sagemaker GPU resources, I fine-tuned the model to classify children vs. adult speech with 98.8% accuracy. I presented the results to my team who plan to integrate this model into educational lessons that help diagnose children's reading difficulties. In the process, I learned a lot about managing large audio datasets, selecting computing resources, and fine-tuning neural networks. For more information, see my Age Classifier project page.

Speech Sandbox March 2025 – December 2025

Curriculum Associates

Last year, I collaborated with another front-end engineer to build a React-based visual audio analysis tool to evaluate performance of automatic speech recognition pipelines. I built an interactive front-end capable of recording and displaying audio, highlighting timestamps over specific sounds, visualizing line graphs of pitch and energy, and showing other types of output. After completion, I presented our work to the engineering department, and the tool became very valuable for other developers to visualize the results of their audio analysis. I hope to extend these skills in my graduate studies in audio analysis and bioacoustics.

Bat Echolocation Call Exploration September 2025

Personal project

I am interested in developing open-source tools to identify bat species by their echolocation calls and support bat conservation. With this goal in mind, I recorded over 115 bat passes in local towns. Then I created a python program to analyze the recordings by extracting acoustic features. I used K-means clustering to fit the calls into groups. Comparing my findings with prior research, I was able to identify a variety of species. The project enhanced my skills in audio signal processing and I'm excited to learn more about how machine learning can enhance bioacoustics research through my graduate studies. For more information, see my Bat Echolocation Calls project page.

HONORS AND AWARDS

Outstanding Computer Science Senior Award August 2022
Department of Computer Science at Framingham State University
- Description: I received this recognition for being a great student in computer science and going the extra mile to help my peers with programming projects

Teaching/Research Assistantship September 2014 – Dec 2015
Department of Psychology at Clark University
- Description: This assistantship involved being a teacher's assistant for one class per semester and conducting research 8-10 hours per week as part of my supervisor's research team
- Value: scholarship covering 100% of tuition with small stipend

Research Assistantship September 2012 – May 2014
Department of Counseling and School Psychology at University of Massachusetts Boston
- Description: This assistantship involved conducting research 9 hours per week
- Value: scholarship covering 50% of tuition with small stipend

Presidential Scholarship September 2006 – December 2009
Ithaca College
- Description: Merit-based scholarship based on high school academic performance and extracurricular activities
- Value: scholarship covering 75% of tuition

TECHNICAL SKILLS

Front end: React, TypeScript, JavaScript/ES6, HTML5, CSS

Data science and Machine learning: HuggingFace, MLflow, Python, SQL, scikit-learn, fastai, librosa

Environments and Tools: git, Jira, JupyterLab, Amazon SageMaker, agile development